

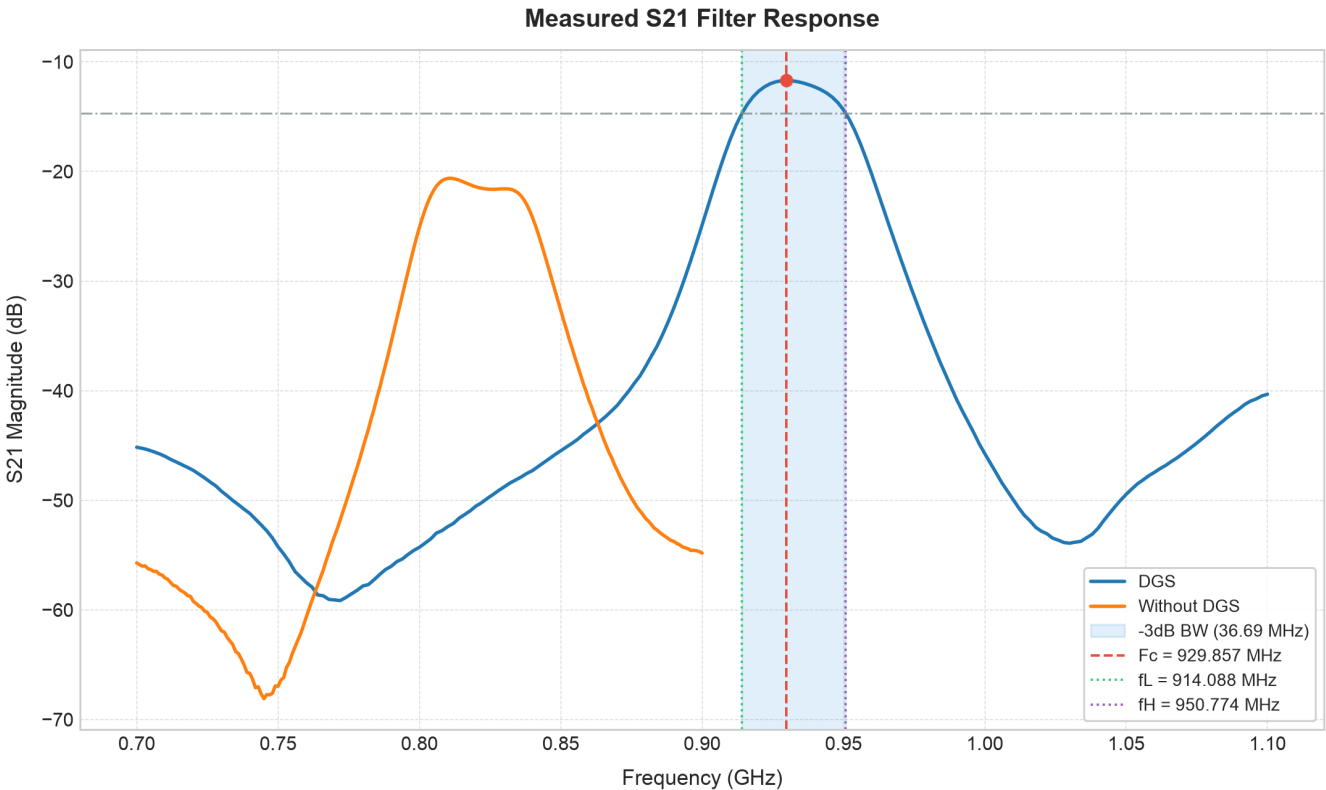
Rohde & Schwarz VNA Filter Analysis Report

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1. Measurement Information

Parameter	Value
VNA Instrument	Rohde & Schwarz Vector Network Analyzer
Active Sweep Points	201
Frequency Range	700.000 MHz to 1.100 GHz
Measured Parameter	S21
Source File	S21_with_DGS

2. Interactive Frequency Response Plot



3. Primary Filter Characterization

Parameter	Calculated Value	Description / Formula
Center Frequency (Fc)	929.857 MHz	Frequency at maximum transmission peak
Peak S21	-11.70 dB	Maximum transmission magnitude
Insertion Loss (IL)	11.70 dB	-Peak S21 (magnitude of attenuation)
Lower Cutoff (fL, -3dB)	914.088 MHz	Interpolated lower threshold crossing
Upper Cutoff (fH, -3dB)	950.774 MHz	Interpolated upper threshold crossing

Bandwidth (BW, -3dB)	36.69 MHz	$f_H - f_L$
Fractional Bandwidth (FBW)	3.95 %	$(BW / F_c) \times 100\%$
Loaded Q Factor	25.35	F_c / BW (-3 dB bandwidth)
Passband Ripple	2.67 dB	Peak-to-peak variation inside passband
Min In-Band S11	-15.34 dB	Best return loss reflection point
Return Loss (RL)	15.34 dB	-Min(S11) in positive dB convention
VSWR @ F_c	1.60	$(1 + \Gamma) / (1 - \Gamma)$ at center frequency

4. Multi-Measurement Comparison

Parameter	DGS	Without DGS
Center Frequency (F_c)	929.857 MHz	810.827 MHz
Peak S21	-11.70 dB	-20.62 dB
Insertion Loss	11.70 dB	20.62 dB
Lower Cutoff (f_L)	914.088 MHz	801.840 MHz
Upper Cutoff (f_H)	950.774 MHz	839.070 MHz
Bandwidth (BW)	36.69 MHz	37.23 MHz
Fractional Bandwidth (FBW)	3.95 %	4.59 %
Loaded Q (F_c / BW)	25.35	21.78
Passband Ripple	2.67 dB	2.96 dB
Min In-Band S11	-15.34 dB	-12.10 dB
Return Loss (RL)	15.34 dB	12.10 dB
VSWR @ F_c	1.60	2.28

DGS Comparative Summary:

- Measured center frequency shift: 119.031 MHz higher (810.827 MHz → 929.857 MHz).
- Measured insertion loss improvement: 8.92 dB (DGS has 11.70 dB vs. Without DGS 20.62 dB).
- Measured bandwidth: 544.36 kHz narrower (-1.5% change, 37.23 MHz → 36.69 MHz).
- Measured fractional bandwidth difference: -0.65% (4.59% → 3.95%).
- Measured return loss improved by 3.24 dB (12.1 dB → 15.3 dB).
- Loaded Q factor changed by +3.57 (21.78 → 25.35).

Engineering Notice & Disclaimer: Reported values are derived from imported VNA measurement data via linear/spline interpolation. S-parameter parameters, Loaded Q, and cutoff boundaries are calculated metrics and should be verified with calibrated laboratory reference standards.